A1 Receive-Coil Amplifier

Hearing-Loop Receiver: Structured Design Review

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Outline

1. Specification & design environment
2. Transfer function & feedback configuration
3. First stage: noise design (g_m, c_{iss}, EKV)
4. Single stage & loop-gain check
5. Dual stage (PMOS pair + NMOS CS)
6. EKV model & pole–zero analysis
7. Frequency compensation (MFM)
8. Conclusions

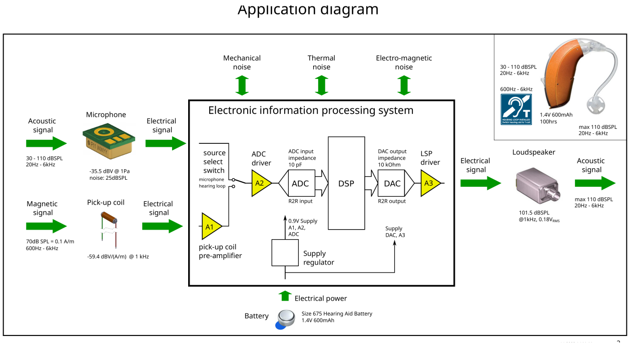
What the review wants

Top-down and *in order*: specify the budget, generate options, then justify every component *before* placing it. Build up to the circuit, don't open with it.

Part 1

Specification & design environment

The hearing loop and where A1 lives



Application diagram, EE4109 HearingLoopIntro

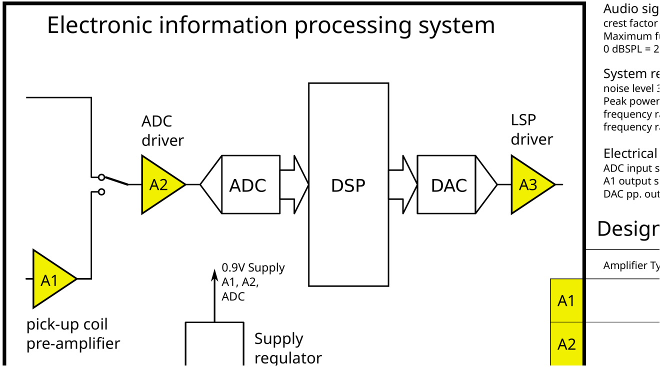
A magnetic *induction loop* broadcasts the audio; a hearing aid picks it up with a coil.

- **A1** = receive-coil pre-amplifier.
- Feeds A2 (ADC driver) ⟶ ADC ⟶ DSP.

Why

Understand the *environment* first, what feeds A1 and what it drives sets the whole specification.

A1 in the signal chain



A1 is the pick-up-coil pre-amplifier

- **Input:** pick-up coil.
- **Output:** source-select switch (mic / loop), then the ADC driver.
- Single 0.9 V supply; 1 mW power budget.

Three questions first

1. What environment?
2. How does it work?
3. What is the budget?

Input of A1: the pick-up coil

The coil output is proportional to the *rate of change* of magnetic flux, it **differentiates**:

$$v_{coil}(t) = -N\frac{d\Phi}{dt} \propto \frac{dB}{dt}$$

- Current in the loop $\rightarrow$ magnetic field $\rightarrow$ coil voltage.
- Open-circuit sensitivity −59.4 dBV/(A/m) at 1 kHz.

Why

If the coil differentiates, A1 must **integrate**, so the cascade is flat across the audio band. That single fact drives the whole topology.

Output of A1: source-select switch + ADC

- **Source-select switch** (mic / hearing loop): needs $V_{mic} = V_{loop}$ at $f_{ref} = 1\,\mathrm{kHz}$.
- **ADC input** is a voltage amplifier: $Z_{in} \to \infty$, draws no current.
- Only loading left = parasitic $C_L = 10\,\mathrm{pF}$.

Consequence

A1 is a **voltage-to-voltage** amplifier driving a capacitor, not a power stage.

Why

Matching $V_{mic} = V_{loop}$ at f_{ref} fixes the required gain magnitude.

How should A1 work?

1. A1 is a **voltage-to-voltage amplifier**.
2. Its transfer is an **integrator** (cancels the coil's differentiation):

$$H(s) = \frac{V_o}{V_i} = \frac{1/\tau_i}{s} = \frac{\boxed{62.4 \times 10^3}}{s}\ \mathrm{V/V}, \qquad \tau_i = 16.06\ \mu\mathrm{s}$$

3. Gain set so $V_{mic} = V_{coil}$ at f_{ref}.
4. **Noise:** weighted output noise $\leq$ microphone floor ($\leq 30\ \mathrm{dBSPL}$).

What the review wants

The order matters: transfer first, *then* noise. And it stays a V–V amplifier throughout, no silent switch to a transimpedance amplifier.

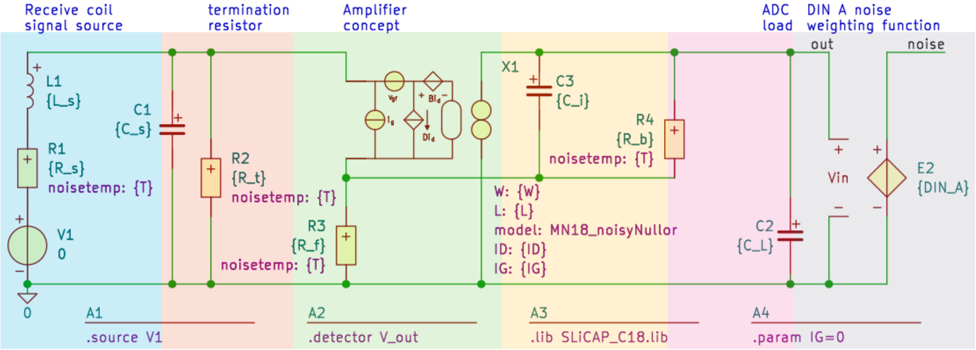
Specification & budget

Quantity	Value	Sets...
Transfer (integrator)	$62.4 \times 10^3 / s$ ($\tau_i = 16.06\,\mu\mathrm{s}$)	gain / $R_f C_i$
Audio band	$f_{min} = 600\,\mathrm{Hz} \ldots f_{max} = 6\,\mathrm{kHz}$	bypass & bandwidth
Output noise (DIN-A)	$\leq$ $10.6\,\mu\mathrm{V}$ RMS ($\approx 30\,\mathrm{dBSPL}$)	input stage
Load	$C_L = 10\,\mathrm{pF}$	stability / drive
Supply / power	1.1 V to 1.3 V / $\leq 1\,\mathrm{mW}$	currents
Coil	$L_s = 120\,\mathrm{mH}$, $R_s = 875\,\Omega$, $C_s = 9.4\,\mathrm{pF}$	source noise / R_t

Why

Every later decision traces back to one line here. The design is graded on *this reasoning*, not on whether the final numbers are perfect.

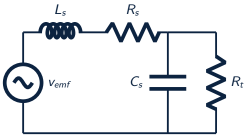
Circuit model: the complete picture



coil source · termination · amplifier (nullor) · feedback · ADC load · noise weighting

The SLiCAP concept model. Each coloured block is represented and justified next.

Representation 1: the pick-up coil



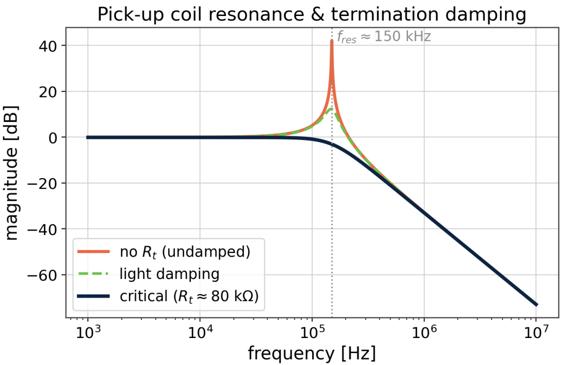
coil source sub-circuit

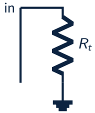
Parameter	Min	**Typ**	Max	Unit
DC resistance R_s	788	**875**	962	Ω
Inductance L_s	102	**120**	138	mH
Sensitivity @1 kHz	-60.9	-59.4	-57.9	dBV/(A/m)
Resonance freq.		**150**		kHz

From the quoted resonance, back out the parasitic capacitance:

$$f_{res} = \frac{1}{2\pi\sqrt{L_s C_s}} \Rightarrow C_s = \frac{1}{(2\pi f_{res})^2 L_s} = \boxed{9.382\,\mathrm{pF}}.$$

Representation 2: the termination resistor





L_s and C_s resonate at 150 kHz; undamped it rings.

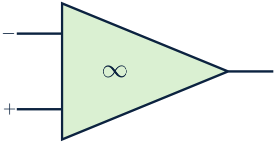
$$Q = 2\pi f_{res} C_s R_t, \quad Q_{crit} = \frac{1}{\sqrt{2}}$$

$$R_t = \frac{Q_{crit}}{2\pi f_{res} C_s} = \boxed{79.97\ \mathrm{k}\Omega}$$

What the review wants

R_t exists *only* to damp the resonance, and it then becomes a budgeted noise source.

Representation 3: the ideal amplifier (nullor)



nullor: $v_{in} = i_{in} = 0$

Sensitivity gives the coil voltage; with $d/dt \leftrightarrow s$:

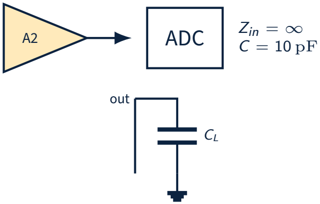
$$V_{ind} = 10^{-59.4/20} = 1.07 \times 10^{-3}\,\mathrm{Vm/A}, \quad V_{ind}(s) = k\,s\,H(s).$$

Convert to Pa and divide by the microphone sensitivity ($-35.5\,\mathrm{dBV/Pa}$):

$$A_{v1} = \frac{V_{mic}}{V_{ind}} = \frac{0.0168}{0.269 \times 10^{-6}} = \boxed{\frac{62.4\times10^{3}}{s}}.$$

With a nullor the transfer is set *only* by the passive feedback, $A_\infty = 1/(sR_f C_i)$; the transistors come later.

Representation 4: the output (ADC load)



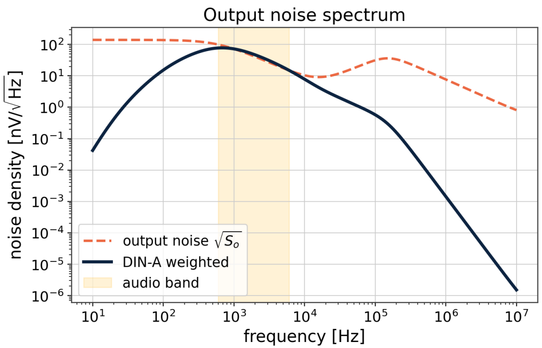
$C_L = 10\,\mathrm{pF}$ is the only load.

The ADC input draws no current ($Z_{in} \to \infty$); only its parasitic capacitance loads A1.

Why

A capacitive load means the output stage must supply charging current at the full-power frequency, and C_L enters the stability analysis.

Representation 5: the noise sources



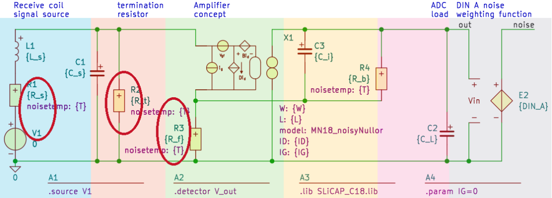
Convert the 30 dBSPL floor to a voltage through the microphone sensitivity ($A_{mic} = -35.5\,\mathrm{dBV/Pa}$):

$$V_{onoise} = 20\,\mu\mathrm{V} \cdot 10^{(A_{mic}+n_{SPL})/20}$$

$$\boxed{V_{onoise} = 10.62\,\mu\mathrm{V}}$$

Each resistor adds $4kTR$; the input device adds v_n, i_n (the noisy nullor).

Noise budgeting



circled: the three budgeted noise sources R_s, R_t, R_f

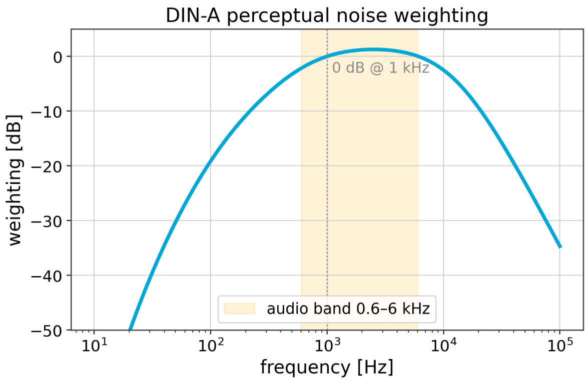
Decide the budget of the *signal source* first:

- Source + termination + feedback: $B_{n1} = 0.4$ of the total.
- Remainder goes to the input transistor, which sets g_m.

$$\text{source share} = \frac{\mathrm{RMS}_{source}}{V_{onoise}} \approx \boxed{0.14}$$

Output-noise RMS $\approx 7.5\,\mu\mathrm{V} \leq 10.6\,\mu\mathrm{V}$. ✓

Noise is judged by ear: DIN-A weighting



Output noise is weighted by the standard DIN-A curve before integrating:

$$\sigma_o^2 = \int_0^\infty |A(f)|^2 S_o(f)\, df\,.$$

- 0 dB at 1 kHz; rolls off outside the audio band.

Why

Weighting makes the budget perceptually meaningful: tight where hearing is sensitive.

Part 2

Transfer function & feedback configuration

First generate the options, *then* choose

Option A, voltage-to-voltage (chosen)

- $Z_1 =$ capacitor, $Z_2 =$ resistor.
- Coil R_s enters only the *noise*.

Option B, current–voltage (transimpedance)

- $Z_2 =$ inductor integrates.
- Coil R_s enters the *transfer*; large inductor.

What the review wants

The prereview lesson: **pick one and stay consistent**. Mixing a V–V amplifier with a transimpedance amplifier between slides makes the plots meaningless.

Decision

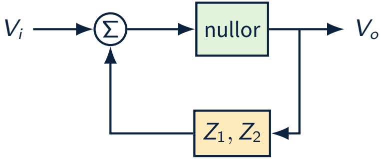
Option A: voltage-to-voltage integrator. Practical components, and R_s stays out of the transfer.

Why feedback? The asymptotic-gain model

Feedback fixes the transfer with passive components, so accuracy follows them, not the transistor.

$$A_f = A_\infty \underbrace{\frac{-L}{1-L}}_{\text{servo}} + \underbrace{\frac{1}{1-L}}_{\text{direct}}$$

- A_∞: ideal (nullor) transfer $= 1/(sR_f C_i)$.
- L: loop gain; servo $\to 1$ when $|L| \gg 1 \Rightarrow A_f \to A_\infty$.



What the review wants

In the review, *explain every curve* of this model, gain, asymptotic, loop gain, servo, direct.

Sizing the feedback network

Bound R_f from two budgets:

- **Noise** ($B_{n1} = 0.4$) $\Rightarrow$ upper bound $R_{f,max} = 7.5\,\mathrm{k\Omega}$.
- **Power** ($B_{i1} = 0.1$) $\Rightarrow$ lower bound $R_{f,min} = 1.4\,\mathrm{k\Omega}$.

$$C_i = \frac{\tau_i}{R_f}, \qquad R_b = \frac{1}{2\pi C_i f_{min}}.$$

$$R_f = \boxed{3.24\,\mathrm{k\Omega}}$$

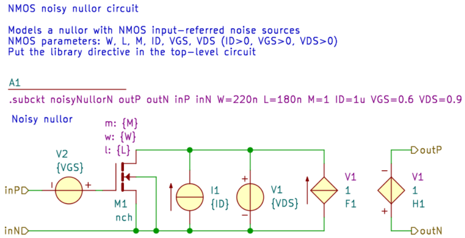
$$C_i = \boxed{4.96\,\mathrm{nF}} \qquad R_b = \boxed{53.5\,\mathrm{k\Omega}}$$

Why

$R_b \parallel C_i$ sets the low corner at $f_{min} = 600\,\mathrm{Hz}$ and gives the DC bias a path, one resistor, two justified jobs.

Part 3

First stage: noise design



Noisy nullor: ideal nullor + the input device's v_n, i_n

Model the controller as a **noisy nullor**.

- Later stages' noise is divided by the first-stage gain ⇒ negligible.
- So the whole budget is spent on *stage 1*.

Why

This is why the first stage is designed *from the noise spec*, before anything else.

The knobs: g_m, c_{iss}, inversion

Input-referred thermal noise of a CS stage:

$$\overline{v_n^2} = \frac{4kT\gamma}{g_m}$$

- More $g_m \Rightarrow$ less noise (more current for less voltage).
- Wider device $\Rightarrow$ more $c_{iss} \Rightarrow$ more input loading.
- **Weak inversion** gives the largest g_m/I_D.

The trade-off

Minimise $g_m \cdot c_{iss}$: enough g_m for the noise budget, lowest c_{iss} for the source.

What the review wants

“Replace the how with the why”: weak inversion is chosen *because* it maximises transconductance efficiency.

Which stage? CS vs CG vs CD

	Gain	Chain (ABCD)	Role
CS	high	**smallest** entries	most *nullor-like*
CG	≈ 1 (current)	—	current follower
CD	≈ 1 (voltage)	—	voltage follower

- The CS chain matrix has the smallest elements $\Rightarrow$ most nullor-like, so later stages barely add noise.
- CG / CD are followers, no voltage gain to suppress later noise.

First stage = **common-source**, in saturation.

From budget to device parameters

1. Find the output noise spectrum $S_o(f)$.
2. Integrate (DIN-A weighted) $\rightarrow$ variance $\sigma_o^2 = (B_{n2} V_{o,n})^2$.
3. Express g_m as a function of c_{iss}.
4. Lowest feasible $c_{iss} \rightarrow g_m, I_D$, then W, L via the **EKV** model.

PMOS input ($g_m \cdot c_{iss}$ optimum):

$g_m = 73\,\mu\mathrm{S}$, $W = 19.1\,\mu\mathrm{m}$

$L = 0.72\,\mu\mathrm{m}$, $I_D = 8.1\,\mu\mathrm{A}$

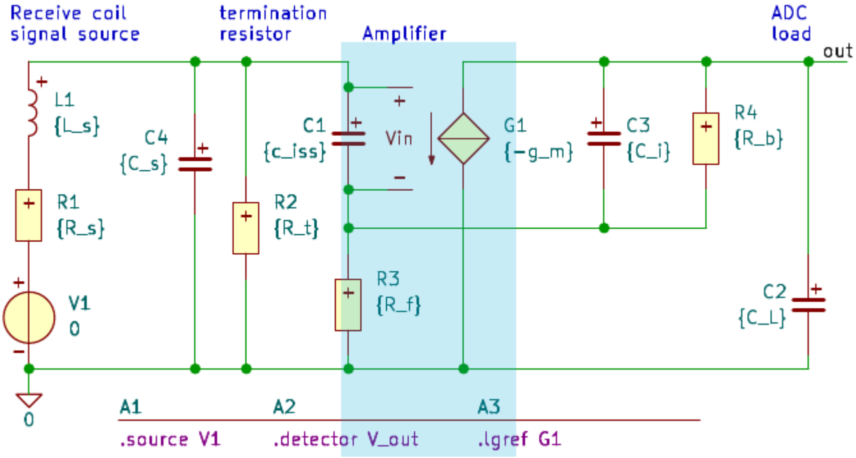
Why

SLiCAP gives the symbolic spectrum; EKV turns the budget into W, L, I_D; ngspice verifies.

Part 4

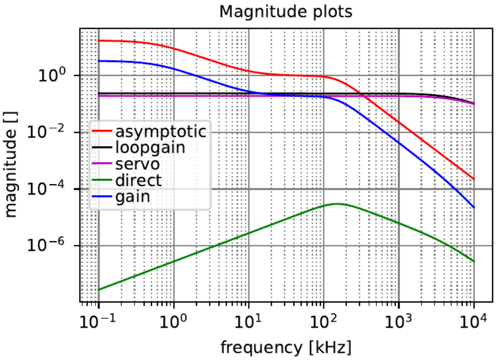
Single stage & loop-gain check

Single stage: the circuit



single amplifier stage ($-g_m$) *behavioural model: coil → termination → one CS controller → feedback → load*

Is a single stage enough?



Loop gain L stays below 0 dB.

A single CS has only 3 usable terminals $\Rightarrow$ a **differential pair** ($\approx 4\times$ current/area).

- Current drive into C_L: ✓
- Voltage headroom on 0.9 V: ✓
- Loop gain $|L_{DC}| < 1$ across the band: ✗

$|L| < 1 \Rightarrow$ servo cannot enforce the transfer $\Rightarrow$ **poor accuracy**. Go to a **dual stage**.

Part 5

Dual stage (PMOS pair + NMOS CS)

Why a dual stage?

Five competing costs: *noise, power, accuracy, stability, area*.

- The **first stage dominates noise**.
- The **second stage dominates loop gain** (accuracy).

Split the job, optimise each stage for one thing.

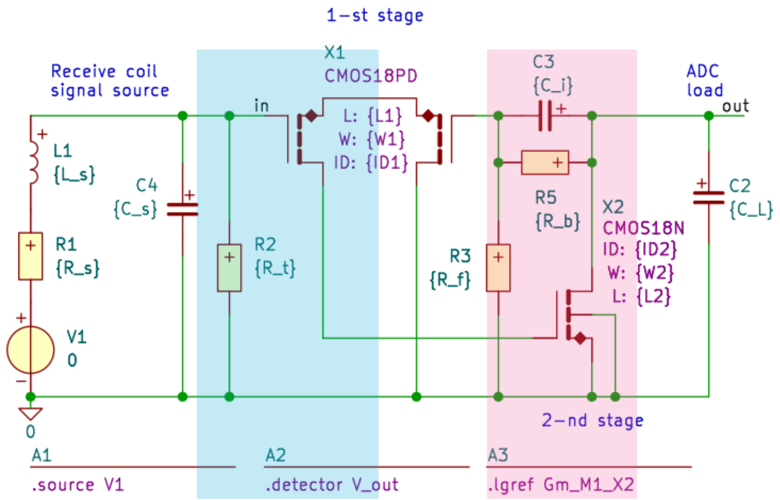
Stage 1: **PMOS differential pair**, low noise.

Stage 2: **NMOS common-source**, high loop gain.

Why

One stage could not give both low noise and enough loop gain; two stages can.

Suggested topology



1st stage: PMOS differential pair 2nd stage: NMOS common-source

First stage: differential pair vs single CS

Differential pair	**Single-ended CS**
+ lower noise	+ less area
+ high CMRR	+ lower power
+ matches the coil's 2-terminal input	+ simple
– more area	– allows CM interference
– higher complexity	– higher noise

The coil is a floating two-terminal source $\Rightarrow$ the **differential pair** wins on noise and common-mode rejection.

Second stage: NMOS common-source

Chosen for what the first stage is *not* optimised for:

- high loop gain $\Rightarrow$ accuracy;
- high current-drive into C_L;
- low power, small area, simple.

Minimum length for speed; width traded for g_m/I_D.

NMOS CS output stage:

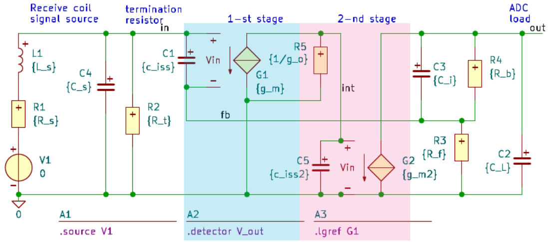
$W = 10\,\mu\mathrm{m}$, $L = 0.18\,\mu\mathrm{m}$

$I_D = 16.5\,\mu\mathrm{A}$

Why

Drive current $I_D \approx V_{i,pp}/2R_f$ must charge C_L at the full-power frequency.

Dual-stage design parameters



behavioural model: g_{m1} (1st) and g_{m2} (2nd) transconductors

$$L_{DC} = \frac{R_f g_{m1} g_{m2} (-R_s - R_t)}{R_s g_{o1} + R_t g_{o1}} = -\frac{R_f g_{m1} g_{m2}}{g_{o1}}$$

g_{m1}, first-stage transconductance, set by the *noise* budget.

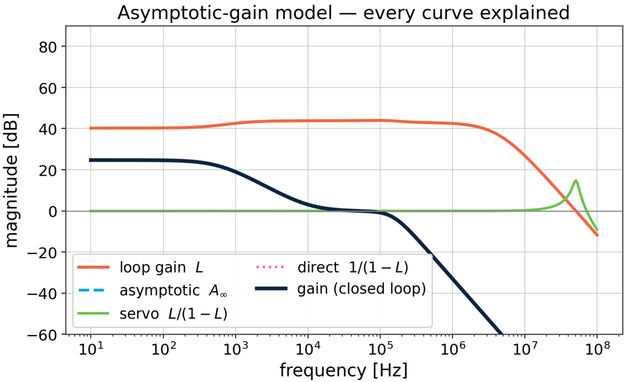
g_{m2}, second-stage transconductance, the extra *loop-gain* factor.

R_f, feedback resistor, set by the *transfer*.

g_{o1}, first-stage output conductance (its output resistance).

$|L_{DC}| \approx 45\ \mathrm{dB}$, enough for accuracy.

Asymptotic-gain model: every curve explained



- **gain**: the closed-loop transfer we deliver.
- **asymptotic** A_∞: ideal nullor target $1/sR_f C_i$.
- **loop gain** L: $\approx 45\ \mathrm{dB}$ at DC, budget for accuracy.
- **servo** $L/(1-L)$: $\to 1$ while $|L| \gg 1$, so gain $\to A_\infty$.
- **direct** $1/(1-L)$: feed-through; negligible.

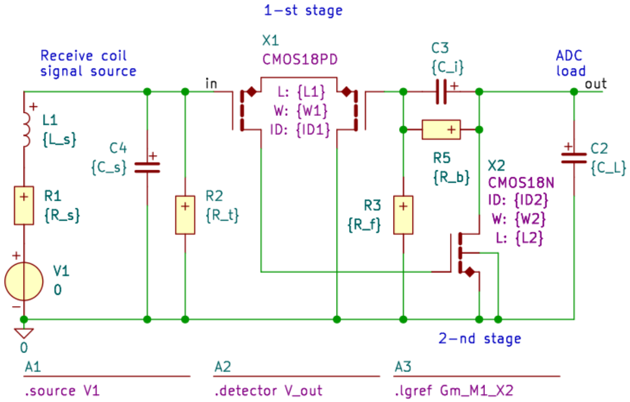
What the review wants

Gain follows A_∞ exactly where the servo ≈ 1; it departs once $|L|$ falls.

Part 6

EKV model & pole-zero analysis

EKV small-signal model



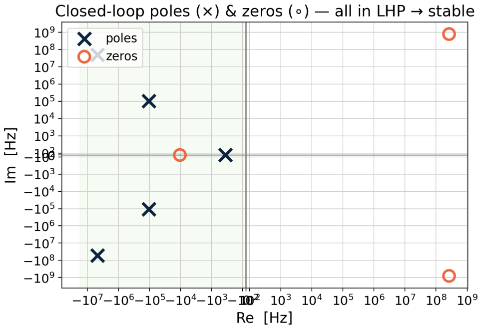
Swap the ideal devices for the **EKV** model:

- realistic transistor-level design;
- includes most parasitic capacitances;
- far more accurate gain & phase.

Why

Now the poles and zeros are trustworthy, so we can check stability.

Pole–zero analysis



	Re [Hz]	Im [Hz]	$\lvert f \rvert$ [Hz]	Q
p_1	−602	0	602	–
$p_{2,3}$	−106.5k	±106.6k	150.7k	0.71
$p_{4,5}$	−4.69M	±50.9M	51.1M	5.44
$z_{1,2}$	+263M	±794M	837M	1.59
z_3	−10.5k	0	10.5k	–

- All poles $\mathrm{Re} < 0 \Rightarrow$ **stable**.
- The 51 MHz pair has $Q \approx 5.4$, severely underdamped $\Rightarrow$ peaking.
- The gain plot follows the poles & zeros, as expected.

Part 7

Frequency compensation (MFM)

Target: maximally-flat magnitude (MFM)

A pole pair with $Q > \frac{1}{\sqrt{2}}$ peaks and rings. **MFM** (Butterworth) sets the dominant pair to

$$Q = \frac{1}{\sqrt{2}} \approx 0.707.$$

- No peaking, fastest flat response.
- Comfortable phase margin.

How to move a pole pair?

Add a *zero* to the loop gain near the offending poles, it injects phase lead and lowers the effective Q.

Why

We shape *stability*, not just gain, a Bode plot is the 0-axis slice of the Nyquist plot.

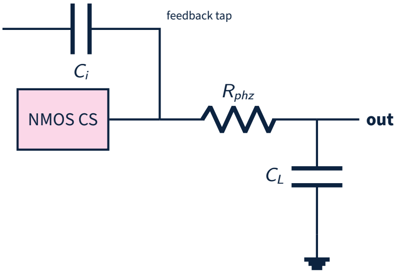
The phantom-zero method

Put R_{phz} in series with the load capacitor: a zero in the *loop gain* with no real pole in the signal path.

$$z = -\frac{1}{R_{phz} C_L}$$

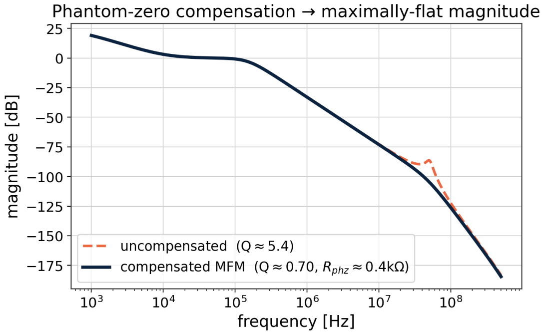
Why this one?

- no extra noise source in the signal path;
- negligible power and area;
- no added energy storage (no compensation cap).



R_{phz} in series; feedback tapped before it.

Compensation result



compensated	$\lvert f \rvert$ [Hz]	Q	
$p_{2,3}$	150.7k	0.71	
$p_{4,5}$	47.8M	0.70	MFM

$$R_{phz} \approx 0.42\,\mathrm{k}\Omega$$
$$Q:\ 5.4 \longrightarrow 0.70$$

- Peak removed; passband untouched; all poles in the LHP.
- Cost: one resistor.

Part 8

Conclusions

Conclusions

The structured path, end to end:

1. Specification & budget from the environment.
2. V–V integrator chosen *after* listing options.
3. Build the model up, block by block.
4. Noise budget $\longrightarrow$ first-stage g_m, c_{iss} $\longrightarrow$ W, L, I_D.
5. Single stage fails accuracy $\longrightarrow$ dual stage.
6. PMOS pair (noise) + NMOS CS (loop gain).
7. EKV pole–zero: stable, one high-Q pair.
8. Phantom zero $\longrightarrow$ maximally-flat magnitude.

What the review wants

The grade is in the *reasoning*: right order, options before choices, every curve and component justified, the how replaced by the why.

$H(s) = \dfrac{62.4 \times 10^3}{s}$, noise $\leq 10.6\,\mu\mathrm{V}$, stable, MFM.

Thank you

Questions & discussion